
CMP6200/DIG6200

Individual Undergraduate Project (FYP)

Final Report

[Safe Shift]

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Table of Contents

Contents

Abstract	3
1.0 Introduction and Context	4
1.1 Problem Statement.....	4
2.0 Review of Existing Knowledge	4
Machine Learning	4
Application Programming Interface (API)	5
Agile Methodology	6
Alerting Systems	6
Location Technologies	7
Microsoft Power Platform	7
Existing Technologies	8
2.1 Critical Analysis and Gap Identification	9
2.2 Project Justification	10
3.0 Project Aims, Objectives, and Scope	10
3.1 Project Aim	10
3.2 Project Objectives	11
3.3 Project Scope	11
4.0 Project Design	11
4.1 Design and Methods.....	11
4.2 Justification of Methods	12
5. Artefact Design.....	14
5.1 Design Specifications	14
5.2 Artefact's Structure/ Architecture / Framework	16
6. Implementation.....	26
6.1 Construction of the Artefact.....	26
6.2 Validation and Testing.....	28
7. Critical Evaluation	32
7.2 Impact	33

8. Conclusion	33
8.1 Limitations	34
8.2 Future Work	34
References	35
Appendices	38

Abstract

Safe Shift is a new innovative solution used to improve user safety and reduce manual rescheduling. Many existing apps in today's world are basic that lack direction and innovation which is what Safe Shift excels in. The report breaks down the details into my development of the idea including the thinking behind the idea, inspiration and project objectives to achieve a solution. These will be explained clearly throughout the documenting highlighting the importance and how critical these are to the development also expanding on existing applications, competitors and potential future work for the project. The report will also expand on existing technologies that have made a difference in today's technology world with Machine Learning, API and location technologies to be documented. Overall, this report should aim to give a detailed overview of the design focusing on the importance of it, approaches and technologies used and many more.

1.0 Introduction and Context

Information Technology is an industry that specializes in innovation as well as productivity, allowing development of new technologies and solutions to solve real-world problems. IT has rapidly transformed nearly every aspect of our lives from communication to entertainment. [Complete I.T. \(2020\)](#). Emerging trends such as AI and quantum computing are set to redefine the industry driving innovation and connectivity. [Globaltechaward.com. \(2025\)](#).

My project is called safe shift it is a prototype consisting of core purposes of providing accurate forecasts and alerting warnings as well as efficient data to users as a mobile app. The prototype focuses on the innovation aspect of IT being a new technology/solution to solve ongoing problems. Most weather apps are criticised for inaccurate data and late warnings for severe weather which damages peoples plans and puts them at risk

1.1 Problem Statement

Data inaccuracy is a common problem most weather apps face today, many of them are criticised by the public for inaccurate or insufficient data making this an ongoing problem. Many severe weather warnings go out late to the public affecting user plans and increasing risk. [Climatecosmos.com. \(2025\)](#). My project aims to solve the problem: false sense of security problems through weather alerts and reducing manual inputs for automation increasing user safety. When severe weather strikes without warning people aren't prepared to take shelter or evacuate safely which is a common risk. My project aims to give alerts to people when severe weather strikes so users can evacuate safely whilst also having optional features such as schedule/calendar integrations allowing users to integrate plans meaning when alerts do happen the product can cancel the plans and reschedule it aligning with the principle of safety.

2.0 Review of Existing Knowledge

Based on my research and papers I have identified, existing technologies are out there close to what my prototype plans to be. The sources have been critical to finding existing technologies and knowledge and talk about current and emerging technologies.

Machine Learning

Machine Learning methods are quite common in terms of using algorithms to predict weather forecasts. [\[Sayor et al., 2025\]](#) uses an application to provide users with real-time weather forecasts utilizing IOT sensors and machine learning algorithms to collect comprehensive data

such as humidity, windspeed and barometric pressure. Climate change increases frequency of disasters like hurricanes meaning data tools such as IOT sensors and machine learning algorithms are crucial for disaster preparedness minimizing loss of life and property [\[Sayor et al., 2025\]](#). Climate change is a major challenge this century as it's not easy to predict when the weather could change due to insufficient data. [\[Guardia et al., 2025\]](#) refers to how specialists used a 3D WebGIS to map precipitation using machine learning algorithms in the region of Palermo (Italy) being developed by JavaScript libraries and using python scripts for activating data connections. API integration, real-time data processing and IOT automation tools are used on technologies that deploy the same idea as mine with built in alerts. [\[Sandhip et al., 2025\]](#) refers to how OpenWeatherMap is used for dynamic mapping and real-time alerts based on location. Machine Learning algorithms have been used to predict crop diseases through real-time weather with different models being used to predict yellow rust algorithms [\[Biljwan et al., 2025\]](#). ML models are used to generate air temperature using historical data fetched from OpenWeatherMap [\[Juneja et al., 2025\]](#). Data fetched from these models have histories of weather events within different meteorological elements like humidity and atmospheric pressure [\[Juneja et al., 2025\]](#). ML Models trained on datasets improved efficiency through metrics such as MSE and RMSE, the metrics presented quantify the difference between predicted temperatures and their actual values providing a clear picture of how accurate the model is fitted [\[Juneja et al., 2025\]](#). Random Forest machine learning algorithms have been developed on Raspberry PI 3 which gets data from humidity and pressure sensors to predict rain [\[Singh et al., 2019\]](#). Different modules are used in which the ML model collects data [from \[Singh et al., 2019\]](#) meaning there are varieties in which it can choose from effectively finding different data. ML models have also been used to map accurate seasonal precipitation forecasts for the Western United States using ML multi-models. [\[Pinheiro and Ouarda 2025\]](#). ML Models like Pangu-Weather and GraphCast struggle to reproduce sub-synoptic weather phenomena leading to forecasts that become overly smooth over time and in some cases ML models lack accurate representation of balances in the atmosphere.

Application Programming Interface (API)

API technologies are perhaps the most common source in terms of gathering accurate weather forecasts. [\[Chadha et al., 2024\]](#) studies aim to create a robust app which provides real-time data to users, in which data is collected from the software OpenWeatherMap and queried using HTTP requests with data fetched in JSON. The data includes current weather conditions, hourly forecasts and daily forecasts with integration parsed by JSON responses stored in a database [\[Chadha et al., 2024\]](#). OpenWeatherMap provides reliable weather data and historical information crucial for a weather app. Forecasts are made by collecting as much data as possible about the current state and understanding atmospheric processes, this forewarns people about future conditions so people can plan accordingly. Once applications are built you get Weather API which uses python source codes to get real-time weather data [\[Dewi and Chen, 2019\]](#). OpenWeatherMap provides an API with JSON, XML and HTML endpoints as well as limited free

user tiers. Historical data access requires a subscription of \$150 p/m in which users request current and extended forecasts with maps showing wind speed, pressure and precipitation [\[Dewi and Chen, 2019\]](#). [\[Hermanth et al., 2024\]](#) refer to how apps leverage technologies like HTML, CSS, JavaScript and integrate OpenWeatherMap API to provide updates for user locations. API in particular processes data using Machine Learning to enhance numerical weather prediction models provided by sources such as NOAA [\[Musah et al., 2022\]](#). Whilst APIs are common, they also produce inaccurate reports and are costly if you need historical information, not all APIs will provide accurate forecasts.

Agile Methodology

Agile Methodology is the most common approach used for problem solving and in this case software/application development. The approach originates from lean manufacturing in the 1940s [\[Spartu, 2010\]](#). The comparisons of agile and mobile application features include documentation, planning, development team size, scale and customer identification [\[Spartu, 2010\]](#). Agile helps companies build the right product and empowers teams to continuously redesign their release to optimize value through development allowing them to be competitive in the market [\[Flora and Chande, 2013\]](#). It focuses more on human interaction over tools and processes considering core values such as interactions, working software, customer collaboration and responsive change [\[Flora and Chande, 2013\]](#). Agile breaks down programming, development and project management into smaller sub problems compared to waterfall methodology where the requirements are defined at the start in which users find it hard to describe what they need the app to do [\[Patil et al., 2016\]](#). Agile assesses each phase giving direction to the project to shape up a final product [\[Patil et al., 2016\]](#).

Alerting Systems

Integrated alert systems are becoming increasingly popular with Smart weather monitoring and alert systems using IOT sensors [\[Rahut et al., 2018\]](#) refers to an app that sends notifications as an effective alert system to warn people about sudden weather changes. Different modules are used on Arduinos and sensors connected by microcontrollers to measure different things. LDR sensor to measure light intensity and ML8511 module using ultraviolet light sensors to warn users of sunburn. [\[Rahut et al., 2018\]](#). System nodes are installed into sensors for data collection and transmitted through a multi-hop network which is processed and sent to people in affected geographic regions via SMS [\[Shiravale et al., 2015\]](#). Radio stations in the United States like NOAA weather radios provide alerts on the radio to users. [\[Collins and Kapucu, 2008\]](#) explains how siren systems are used across the Midwest region of United States alerting communities of impending weather disasters with a loud blare. NOAA weather radios broadcast continuous weather forecasts from nearby offices which activates whenever an alert is sent out however, not many people own a NOAA radio meaning they miss out on potential weather alerts for

severe conditions so if a transmitter loses power other disaster alerts may not be sent out in a timely manner [\[Collins and Kapucu, 2008\]](#). IOT and Machine Learning technologies have been used to warn users of severe conditions using hardware sensors to measure aspects like air pressure, humidity and carbon monoxide in the atmosphere [\[Durrani et al., 2019\]](#). These algorithms were prepared using NARXNET technology to check the predicted data to see if it fulfils the conditions of an alert, if that is the case it generates alert for disasters such as heat stokes and rainstorms [\[Durrani et al., 2019\]](#)

Location Technologies

Location-based services aim to provide information to users based on geographic area at any time [\[Dao et al., 2002\]](#). Pinpointing specific location is challenging which is why non-GPS solutions use COO (Cell of Origin). COO is straightforward as it uses cell identification for approximate location of the caller [\[Dao et al., 2002\]](#). LBS is a major requirement for a GPS locator as this allows users to get their own current location and view on google maps. Locations are calculated with built in GPS with coordinates on maps using third-party API called MIDMAPS which is developed through JAVA 2.0 and uses MySQL for database hosting [\[Chandra et al., 2011\]](#)

Microsoft Power Platform

Microsoft Power Platform is an increasingly growing cloud solution used by many businesses nowadays being used widely across various industries and departments including IT, business operations, sales and marketing. Leading organizations worldwide in tuning into Power Platform to revolutionise their operations with over 33 million monthly active users driving innovation and transformation [\[Lutz, M. 2024\]](#). Cineplex, a media and entertainment company based in Canada have managed to save more than 30,000 hours annually due to the automated key processes on the platform, as well as Insurance companies such as Community Fibre reducing business process handling times by 85% using Power Automate. It is clear to see with these big companies that Power Platform will continue to grow soon.

Microsoft Power Platform offers an engaging and a straightforward way to use data to help run your business combining capabilities with the 3 solutions Power BI, Power Apps and Power Automate to help businesses gain insights into data through dashboards and reports, leverage data insights to drive innovation and improvements through applications and automate workflows to save time and drive efficiency.

Power BI

Power BI is a tool primarily used for data visualisation and reporting. As of February 2017, more than 200,000 organisations across the world are implementing Power BI due to it being a simple solution that automatically build reports based on analysis of data [\[Krishan, V., 2017\]](#). This means it reduces human errors in terms of calculating and analysing which can lead to unverifiable research. Visualisation tools are used which range from coloured maps to single numbers helping present data in a way that provides context and insights. When reports are published the report can be included in dashboards which show a 360-degree view of the data enabling businesses to keep their most important metrics in one place [\[Krishan, V., 2017\]](#).

Power Apps/Low-Code/No-Code

Microsoft Power Apps is a low-code, no-code tool designed for organisations to build custom applications suiting their business requirements. Organisations worldwide are starting to use Power Apps to address large-scale business challenges and save time than if they were to opt with heavy coded solutions. Microsoft has shifted its focus to now become the most comprehensive low-code application development strategy with approximately 92% of Fortune 500 companies using Power Apps each month allowing them to provide customers, partners and employees with streamlined and top-tier solutions [\[Korada, L., 2022\]](#). Modern low-code no-code solutions have many benefits over manually writing lines of code with programming languages, these benefits include faster development with cloud solutions where anyone without coding or technical experience can develop complex solutions without having to write complex lines of code. This benefit alone makes the solution more popular for organisations as it provides a quick advanced level solution saving time than hard coded solutions [\[Sufi, F., 2023\]](#). Low coded or no coded platforms enable quick generation and delivery of applications with minimum effort of coding as it is the fastest and cheapest way of developing software [\[Waszkowski, R., 2019\]](#).

Power Automate

Power Automate is a tool that helps organisations build multi-stage workflows that automate repetitive tasks saving time and increasing productivity. Much like Power Apps no relevant coding experience is needed making it suitable for users of non-technical backgrounds to scale a solution. One of the largest factors Power Automate is increasingly popular is due to the fact the automated workflows can be directly connected to Power Apps through connectors such as SharePoint, Excel or Lists automating data on Power Apps through workflows. Power Automate can provide maximum advantage when it comes to improving personal productivity such as setting reminders, notifications or data updates [\[Wadiwala, R and India, S., 2019\]](#)

Existing Technologies

Similar apps like AccuWeather are known to be some of the best out there but lack alerting systems. AccuWeather allows push notifications which can be turned on/off by users, these notifications are location specific instead they're radius specific sending alerts within 10 miles of

a user's location. NOAA radio stations provide alerts for anyone who has them installed but not for users that don't have them increasing risk of safety for them. Disagreement amongst people is common when receiving multiple weather updates from different sources. In October 2016 historic and destructive winds were forecasted for 15th October with initial warnings being sent on Wednesday and by late Friday the chance of it happening was decreasing [\[Burgeno and Joslyn, 2020\]](#).

2.1 Critical Analysis and Gap Identification

Based on the sources identified existing technologies have been used to create similar applications involving machine learning and API integrations but there are a lot of faults, some of these faults include the inaccuracy of data forecasting. Whilst recent technologies and projects have shown good progression with these integrations Apps like AccuWeather have been heavily criticised for inaccuracy, as well as the push notifications not working as well as it should be. Many weather applications use notifications, but these can be turned on or off even displaying radius data or insufficient data. The gap identification being using effective alert systems to alert users about potential severe warning in their geographical area instead of notifications 10 miles away.

While Machine Learning algorithms are great the downsides to them are that they require high quality and extensive datasets also having critiques in ML models being expensive requiring significant investment in infrastructure. API integration has its limitation which include rate limiting. Exceeding these limits result in additional charges or denial of service. Even with all the negatives of existing methods the clear gap I have identified is regarding in alerting systems while we can expect some sort of data inaccuracy the gap involving safety remains an issue. NOAA radio stations provide real-time forecast accuracy though broadcasts however, the downside to this technology is that not everyone in the US owns a NOAA radio station so getting severe weather alerts to them is seemingly a challenge. Agile Methodology is a commonly used approach where people break down larger parts into smaller sub parts to create a valuable outcome, but this comes with costs such as unstable planning and misinterpretation of feedback from clients.

Power Platform has many advantages such as speed and easy access for non-developers but compared to hard coded solutions they lack customisation, tools and integrations due to being drag and drop icons under a plan as opposed to hard coded solutions allowing a wider range of customisation features due to there not being any plans. However, many businesses are transitioning into Power Platform making it a popular solution for application development as not everyone will be able to code.

2.2 Project Justification

The justification of my project is clear whilst there are many advantages in which applications use these technologies, they haven't solved a critical gap regarding safety which is what my project intends to do. Based on the critical analysis the gap is the use of alert systems for safety. This is a common issue, and similar technologies only utilize push notifications as an alert system. This makes my project relevant as the intended prototype will build on this problem using alert warning systems based on a user's specific location instead of nearby locations. This is what many existing apps lack as they use location technology for severe alerts based on the user's nearby city for example 10 miles. Push notifications can be turned on/off meaning users can miss severe weather forecasts, my app won't allow that as it will directly alert the user with a beep like how they use systems in government tests for threats etc. This is a gap that has not yet been explored and is a core purpose to why it is relevant, it improves safety and allows users to plan accordingly based on alerts with scheduling and cancellation systems rescheduling activities based on the alerts. Aside from using the location and alerting system technologies the main development will consist of existing technologies such as machine learning and API integration however, my project aims to use these to pull accurate efficient data instead of inaccuracies some applications have. As a summary to why my project is relevant, many apps have forecast inaccuracies and fail to implement warning alert systems to alert people of severe weather causing tragedies such as loss of life.

3.0 Project Aims, Objectives, and Scope

3.1 Project Aim

The aim of my project is to develop an application to solve the false sense of security problems through weather apps, by using integration tools such as API keys or connectors and tools such as VS Code and Power Platform to pull real-time weather data and location-specific technologies to create customized weather alerts based on users coordinates whilst also having rescheduling systems to update a user's plans based on weather conditions. The weather will send alerts notifying if it is safe for the individuals to go ahead with their plans or not, this innovative idea can contribute to user awareness and safety.

3.2 Project Objectives

- *Research into existing technologies / Problem identification*
- *Evaluate tools and software for prototype development*
- *UI/UX designing / Model development*
- *Application development / Feature integrations*
- *Test solution / Gather user data / Evaluation*
- *Deploy and publish application*

3.3 Project Scope

This project will focus on software development and integration methods. It will include software's such as Power Apps to help build the UI front-end design as well as the built in API connectors to pull real-time data. The project doesn't include any hardware materials to pull data as it is strictly software based. The prototype will include many features when completed such as location-specific coordinates, data forecasting and data management however, it won't include advanced or complex data formats as this makes it unrealistic for me and the project. Overall, the prototype focuses on user interaction using innovative features and not focus on complex coded integrations. Hard coded solutions give you more flexibility in customisation but for me using platforms such as Power Apps can save time and stress equally contributing to an effective solution.

4.0 Project Design

4.1 Design and Methods

Objective 1

Design & methods link to the project objectives and show how I will achieve each one. Objective 1 focuses on data collection before development as well as research into existing applications. Gathering user data early will help my find the problems user's face with so that I can develop an app suited to them. From gathering the data and researching into existing solutions I can narrow down specific problems and begin to create a solution.

Objective 2

After research and user data has been collected the next step will be to start evaluating tools and software's used for the development of the application. The most effective way for me to do this is by researching and testing a lot of application-based software's and using their features to find which is the best fit for me, these tests will include coded and low-code solutions, costs, performance, speed and other factors.

Objective 3

After choosing the right software, the development stage begins. The user data will be the main process into the development stage as its for them giving them what they want. This objective can only go ahead of objectives 1 and 2 are met successfully.

Objective 4

Carries on with the development stage from objective 3, this time focusing on feature integrations such as the calendars, connectors/API's and location technologies rather than the UX.

Objective 5

This objective will be conducted after the prototype is done to my liking, this stage focuses on user data after the prototype so they test out the features etc and I can gather the responses and evaluate on the things they liked and didn't like so I can improve on them for future work. Data for this stage will be conducted through online forms

Objective 6

Once happy with the application it can be deployed and maintained for all users to interact with.

4.2 Justification of Methods

Objective 1

Qualitative forms will be used rather than quantitative numerical forms as it allows me to gather specific knowledge of user's thoughts and problems through descriptions rather than numbers on a scale. For an application it's better to have a qualitative form as it gives you detailed insights into users' problems making it easier to develop an application to suit their needs as they are describing their problems and writing what they want and expect. Researching into existing technologies gives me insights into what features they use, so I can understand the methods and

identify potential strengths and weaknesses within those solutions to give me a better foundation when developing.

Objective 2

Evaluating tools and software's gives me the best knowledge and preparation for development, knowing what each software is capable of by testing their features and systems it gives me deeper knowledge so I can vary my decision based on the ones I have used and the ones I'm better than others at. Cost, features and more come into consideration when choosing but overall, it is to see which software is the right fit for my app and which is best for targeting user needs whether hard code or low code.

Objective 3

Development starts here due to all the data gathering and research being collected. This allows me to focus on the app knowing I already have the data needed and software chosen to tailor it to users needs. After weeks of identifying the software the UI/UX stage begins on it so that the development process can be smooth and efficient.

Objective 4

Carries on with development using the data collection and chosen software. Techniques and features implemented into the software to make the prototype relevant to what intended to do.

Objective 5

Final data gathering allows me to collect user data from the same or different users once the prototype is finished. This will be in the form of a qualitative form which users test the app and provide their feedback on it allowing for me to assess and evaluate what went well and what didn't go so well. The justification for this is crucial for future work allowing me to plan for the feature and design an updated version focusing on their bad experiences if they had any.

Objective 6

Once data is collected the app will be deployed and published anyway so I can receive feedback from the public analysing reviews focusing on future developments and updates for the app. The app will be accessible and available to everyone reaching a wider audience than it is currently.

5. Artefact Design

5.1 Design Specifications

This following section will aim to provide overview of the requirements and specifications of the prototype. When initially planning my prototype months ago, I had a vision of implementing alert or notification systems into existing weather apps allowing the system to reschedule a user's plan based on weather severity. The idea might not seem logical but I believe any bit of innovation whether creating an entirely new app or implementing new features into existing solutions play a pivotal role in business success. Although the project is a prototype meaning it is essentially an unfinished or first version of a high-end application there are still many requirements involved. Some of the basic requirements include navigation, modern UI and data sources. In terms of what the artefact must do, it must show valid weather data and conditions for any city the user chooses, it must show the users home location and coordinates and most importantly it must show valid data along with optional coded data included Celsius and Fahrenheit. The application must be accessible to users and should have smooth navigation between pages. Below is a more in-depth guide to the prototype requirements, features and constraints.

Requirements	Why	Must Do	Why	Constraints	Why
Modern UI	For a cleaner look to the app, enhancing user experience	Show Weather Data and Forecasts	The main data of the app, it must show this for the rest of the app to follow on, allows users to check forecasts and make decisions	Limited Forecast Data	No code solutions like power apps allow 2-day forecasts, this makes it harder for rescheduling across a week
Basic innovative solution, no premium modes	No complexity, based on data users want an innovative solution contributing to safety and not risk payment	Navigate between pages	Must do this because it allows users to access other pages and use the app fully	Record Limits	Power Apps allows 500 records but can increase to 2000 which affects performance, shouldn't affect me as I won't have more than 500

					schedules for a prototype
Navigation	Improved user experience and enhanced engagement allowing flow between pages	Update and find user locations	Allows users to know their exact coordinates and have weather automatically shown based on where they are, wherever they change to, location will be updated so they know where they are	Formula Language Complexity	Although low code solution certain parts require coded formula which can prove challenging to implement
Data	Cleaner look to the system, working prototype, allows improved decision making and regular updates based on city or coordinates	Text Input	Must allow users to have access to enter any city in the world and have the weather forecasts for them, enhancing user experience	Limited Customization	Compared to hard coded solutions, low code has lower customization features as its more used for quick scale business applications, this makes it harder to have full control over the app
Clean Text, Size and colours	Increased readability and enhanced user experience, having the right colours makes it accessible for anyone with varying levels of focus	User Alerts / Notifications / Schedules	Allows the app to send notifications to users based on weather notifying or alerting them of severity. Will also reschedule plans if the weather data is severe, if not it stays the same.	Licensing Costs	The effect of cost has an impact of development as users will spend money trying to design a solution

Table 1 – Design Specifications

5.2 Artefact's Structure/ Architecture / Framework

For this prototype 2 approaches were used, both in which show how components interact with each other and technologies used. The main structure used for the artefact was Microsoft Power Apps as it provided stability and customization with speed and optimization. The choice of this structure allowed components to interact with each other smoothly. For example, buttons can easily be navigated through this technology with buttons saving data and allowing user inputs, the structure allows the components to do this based on the code and programming involved. Components interact with click or text inputs on most applications which is the same approach my project follows. The first approach later switched to power apps was done through VS code and Python Programming which consisted of components such as navigation bars, text input bars and more. Technologies used in this approach included external third-party Weather API tools to gather accurate weather data and the main software itself VS Code. The components are the driving force of the app and as previously mentioned include icons and the design but also include the logic such as data storage and access. The prototype focuses on many structures which include the data, interactions, integrations and logic. Below are tables breaking down the structure and architecture of the prototype explaining the components and their interactions, technologies used in the process of development and why they were chosen as well as the algorithms and materials used.

Key Components	Interaction
Buttons	Users click buttons on the screen allowing them to navigate between pages, prototype includes clear icons which users click navigating to specific pages
Lists/SharePoint	When data is stored/saved it is saved to a backend list, showing all user data to the developer keeping track of inputs
Screens	Main component of the app, screens allow the user to familiarise themselves, buttons allow user to change pages
Data	Users manually enter data through text input components in which it is saved to a backend data source being a list
Processing	Main component, when data is created by the user automation logic is applied to process, read and store the data allowing changes to happen based on user inputs.

Table 2 – System Architecture

Technologies	Why
Microsoft Power Platform / Power Apps / Power Automate	Easier Development, Rapid Deployment, no technical knowledge required, popular source of application development, speed data integration. Data can be automated in the back without manual inputs through flows allowing for functionality and changing features
API/Data Connectors	Needed for weather data, showing users forecasts for any city or country around the world, makes the app functional and can solve user problems, data connectors allow various data to be implemented into the app linking different technologies such as lists, SharePoint and Dataverse with data being stored when connected to the app and user interacts
Geolocation/GPS	Allows users to see exact coordinates, nearest city and distance from equator, optional features make the app feel more modern, stops confusion of where they are as coordinates are pinpoint, linking the weather forecast to their area
SharePoint	Backend data source used for storing and managing user data entered through the canvas app, allows developer to see user responses. Can then make decisions to reschedule or cancel user activities based on data through API and connectors

Table 3 – System Architecture Continued

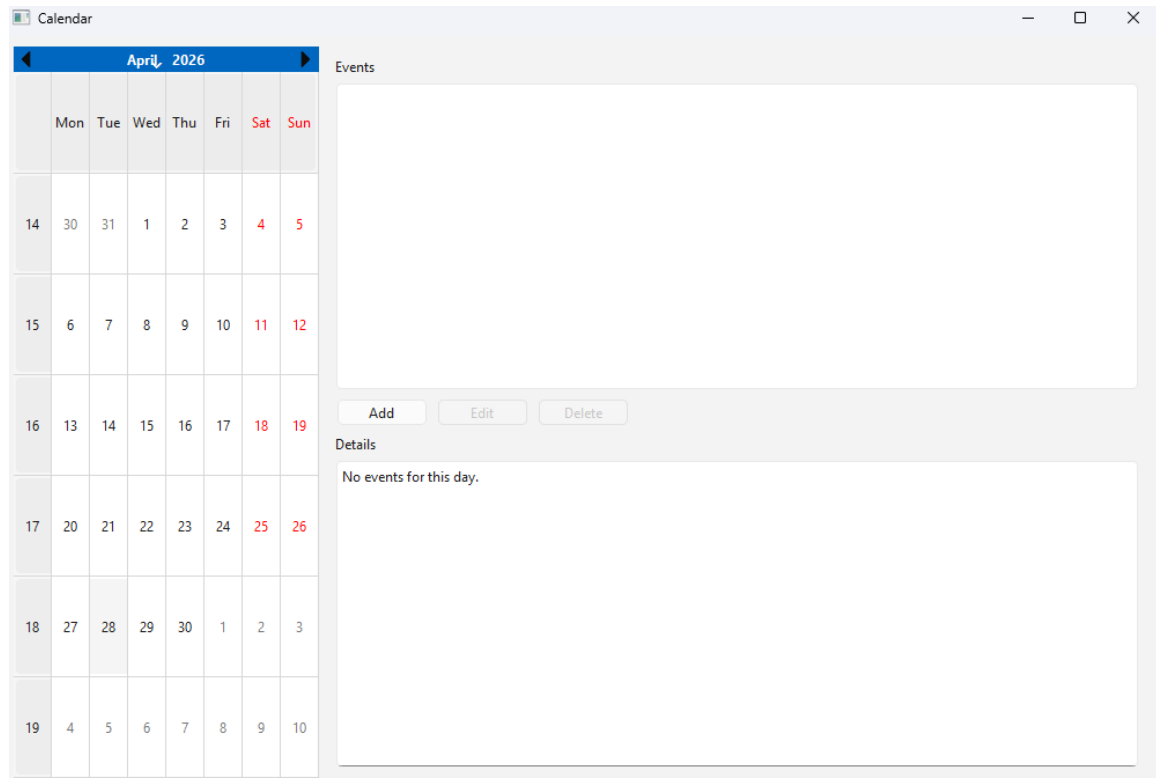


Figure 1 – VS Code Calendar

Figure 1 shows my calendar app from my initial approach which was full stack coding using Python and Visual Studio Code. This app is basic but uses core programming knowledge for the UX and functionality of it, as you can see from the figure days and months are displayed like any generic calendar with the add option, when users interact with the add option by clicking on it, it allows the users to add schedules or information for that specific day which gets saved automatically on the app. Not much functionality was used for this app as its rather just a failed attempt of me trying to bridge 2 apps (Weather + Scheduling) into 1 system. Other components of this app include arrows at the top allowing users to navigate between months and years allowing stable flexibility to a minimal application.

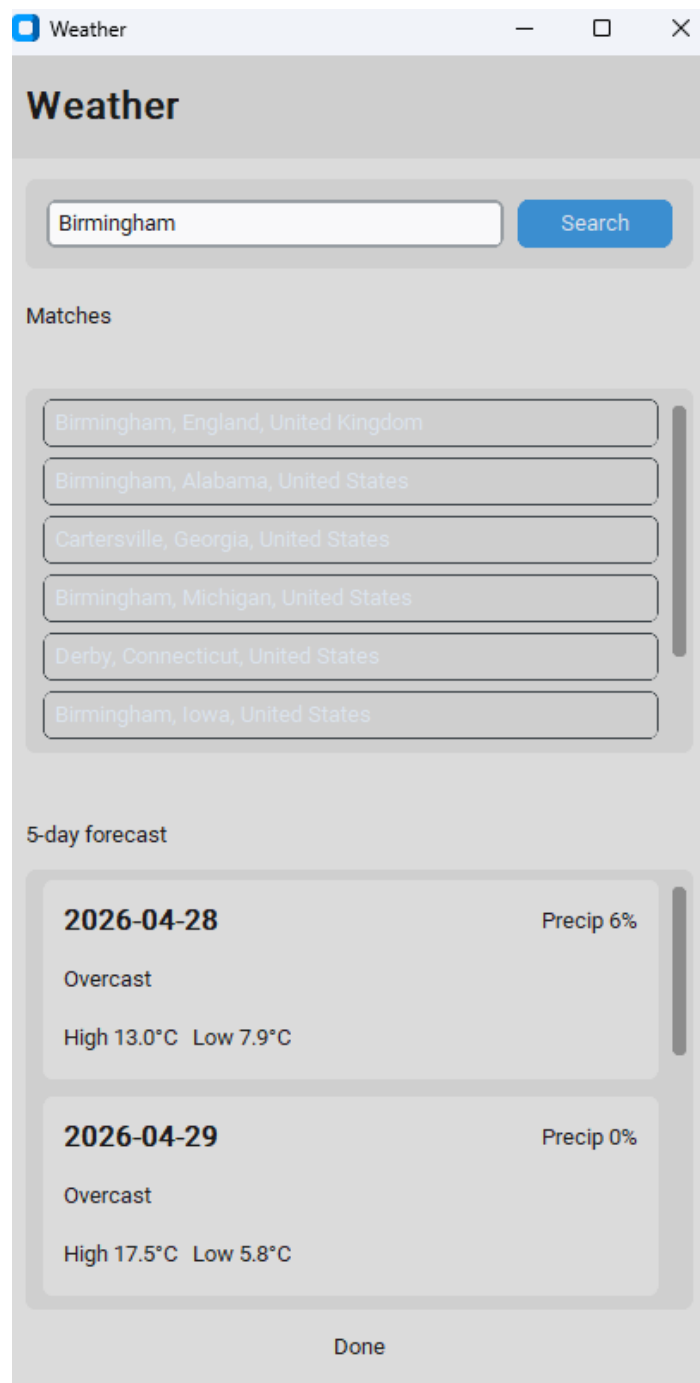


Figure 2 – VS Code Weather App

Figure 2 shows my weather app created on Visual Studio Code, as you can gather it is a basic weather app that does what any competitor app does, displaying weather data through the boxes providing information such as highest and lowest temperatures as well as percentage of rain. This is done by the user interacting with the search bar right at the top of the screen, by users entering any city the app displays data and suggests other cities as recommendations.

These are the key components of the app doing what is expected of a typical weather app, compared to the calendar app, it has a lot more components such as interactive buttons when choosing a city and etc. The initial idea was to build the 2 apps into 1 through full stack but after that became difficult, I created 2 minimal apps with a lack of components and technologies to show the progression of my app during my new approach.

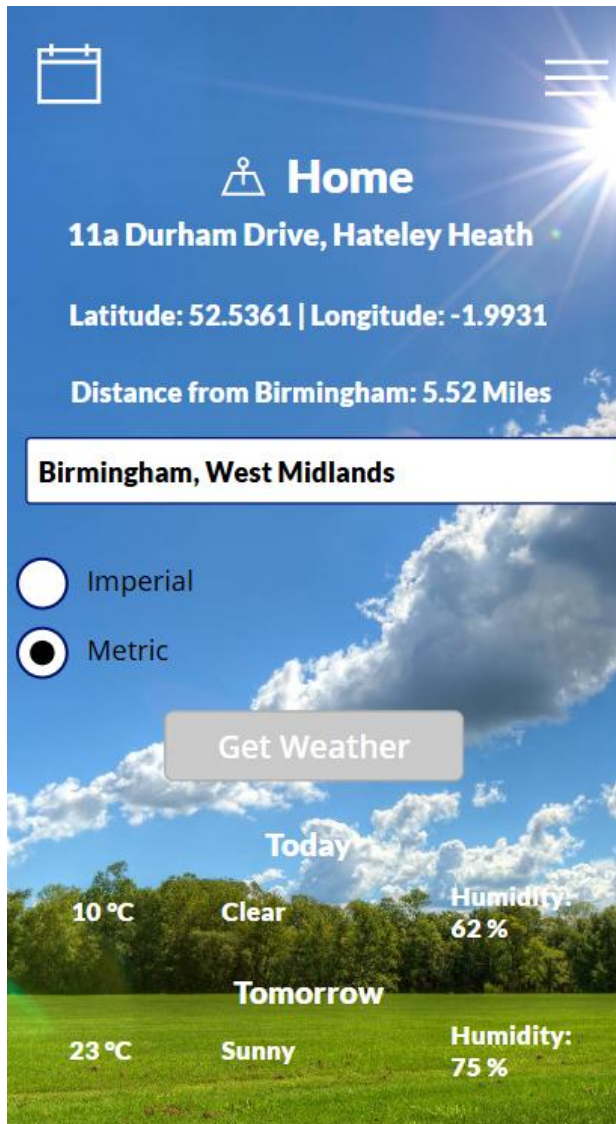
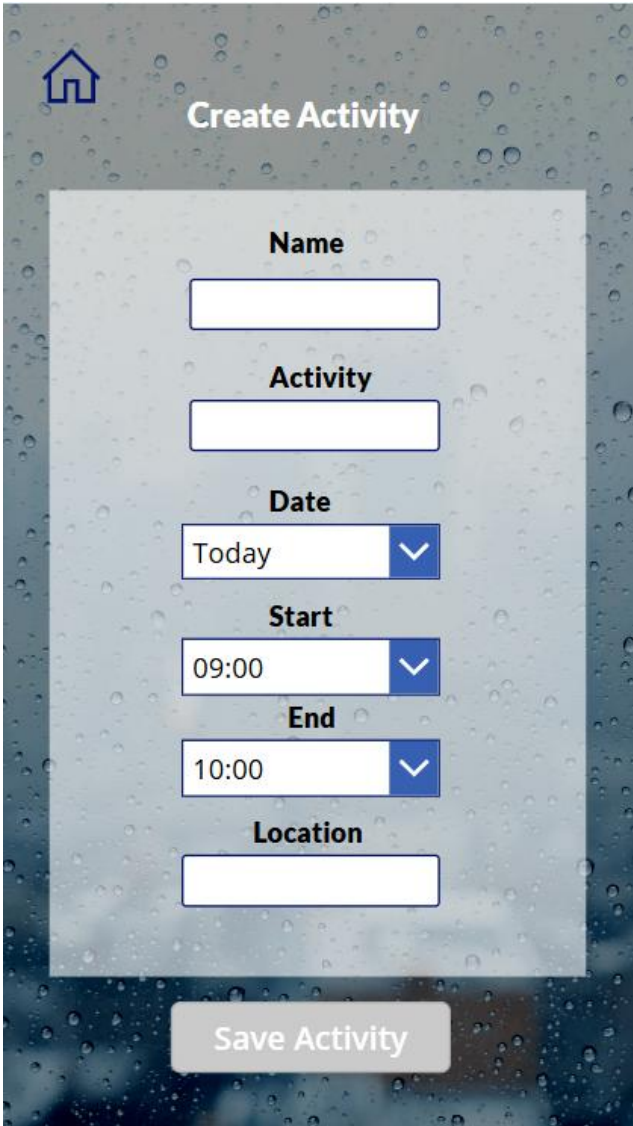


Figure 3 – SafeShift Home Screen

Figure 3 shows my design in Power apps as it blends the 2 previous VS Code designs into one solution. The system uses automation as we can see here from the screen the users' exact coordinates and location are displayed underneath home as well as their distance from Birmingham. These components interact based on a user's movement as it uses GPS technologies to change a user's location based on their movements, something not many weather apps do. The text input box in white allows users to find weather for any city based on

what they type displaying the city and state. Imperial and Metric options add to user experience allowing users to select which units they prefer to see the weather in. Get Weather is a component which lets users then click after they enter their preferred city and units displaying today and tomorrow's weather data below which pulls data from external API connectors. In summary components interact through typing and clicking just like a typical app with automatic features enhancing user experience and giving the home screen a more modern feel.



 **Create Activity**

Name

Activity

Date
Today 

Start
09:00 

End
10:00 

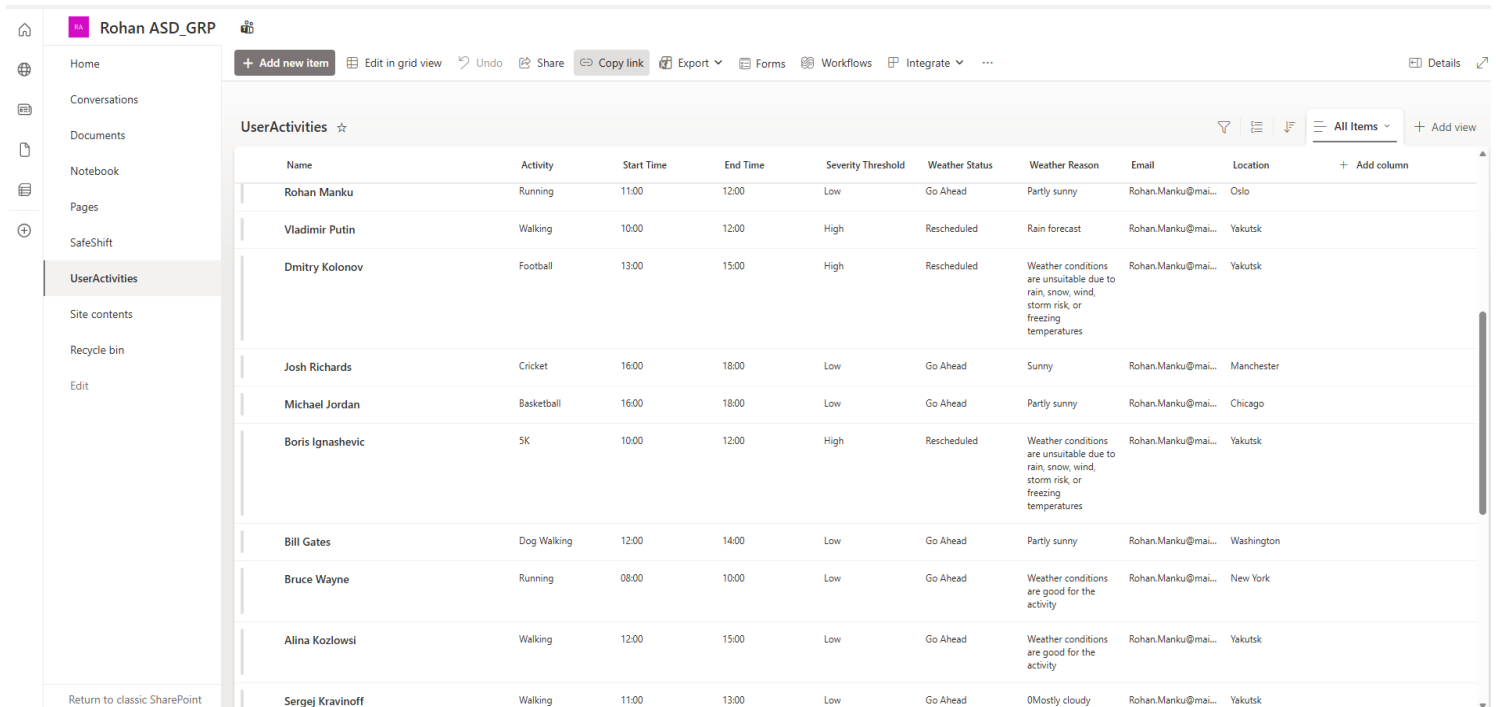
Location

Save Activity

Figure 4 – Page 2 SafeShift

Figure 4 shows page 2 of my prototype, this page allows users to enter an activity with dropdown menus for date, start and end with a text input box much like the home screen letting users choose a specific location where their activity is taking place. The components all interact with each other through typing and clicking allowing seamless interaction and enhanced user experience. User experience is important as its defined as a person's perceptions, such as how

well it works, how they feel about it and how well it fits into the context they are using it ([Hassan and Dahlan 2013](#)). And based on the interactions of components from both the activity and home page, potential users will have a full understanding of the concept and how the app works due to the easy navigation and text inputs.



Name	Activity	Start Time	End Time	Severity Threshold	Weather Status	Weather Reason	Email	Location
Rohan Manku	Running	11:00	12:00	Low	Go Ahead	Partly sunny	Rohan.Manku@mai...	Oslo
Vladimir Putin	Walking	10:00	12:00	High	Rescheduled	Rain forecast	Rohan.Manku@mai...	Yakutsk
Dmitry Kolonov	Football	13:00	15:00	High	Rescheduled	Weather conditions are unsuitable due to rain, snow, wind, storm risk, or freezing temperatures	Rohan.Manku@mai...	Yakutsk
Josh Richards	Cricket	16:00	18:00	Low	Go Ahead	Sunny	Rohan.Manku@mai...	Manchester
Michael Jordan	Basketball	16:00	18:00	Low	Go Ahead	Partly sunny	Rohan.Manku@mai...	Chicago
Boris Ignashevich	SK	10:00	12:00	High	Rescheduled	Weather conditions are unsuitable due to rain, snow, wind, storm risk, or freezing temperatures	Rohan.Manku@mai...	Yakutsk
Bill Gates	Dog Walking	12:00	14:00	Low	Go Ahead	Partly sunny	Rohan.Manku@mai...	Washington
Bruce Wayne	Running	08:00	10:00	Low	Go Ahead	Weather conditions are good for the activity	Rohan.Manku@mai...	New York
Alina Kozlowski	Walking	12:00	15:00	Low	Go Ahead	Weather conditions are good for the activity	Rohan.Manku@mai...	Yakutsk
Sergej Kravinoff	Walking	11:00	13:00	Low	Go Ahead	Mostly cloudy	Rohan.Manku@mai...	Yakutsk

Figure 5 – SharePoint Data

SharePoint data relates back to Figure 4 when users interact with the components on that page via type, select and click the activity is saved and then stored to a SharePoint list where me, the developer has access to all of the users data, documenting the weather conditions, if the weather conditions are suitable for the activity to go ahead, if its not etc. It displays the severity of the weather based on the location the user chooses through page 2 and notifies users if they should proceed with the activity or not. Rescheduled status is applied based on checking tomorrows forecast before the system decides to reschedule the activity or if it should go ahead based on good weather. Weather Reason tells you why the activity goes ahead or is rescheduled contributing to user safety and an innovative solution. SharePoint enables organisations to deliver business applications in an effective manner and is seen as a critical business tool allowing organisations to share knowledge, manage data and content ([Khumalo and Mearns 2019](#)) which is exactly why SharePoint is appropriate for this solution allowing me to share and manage data being the user responses.

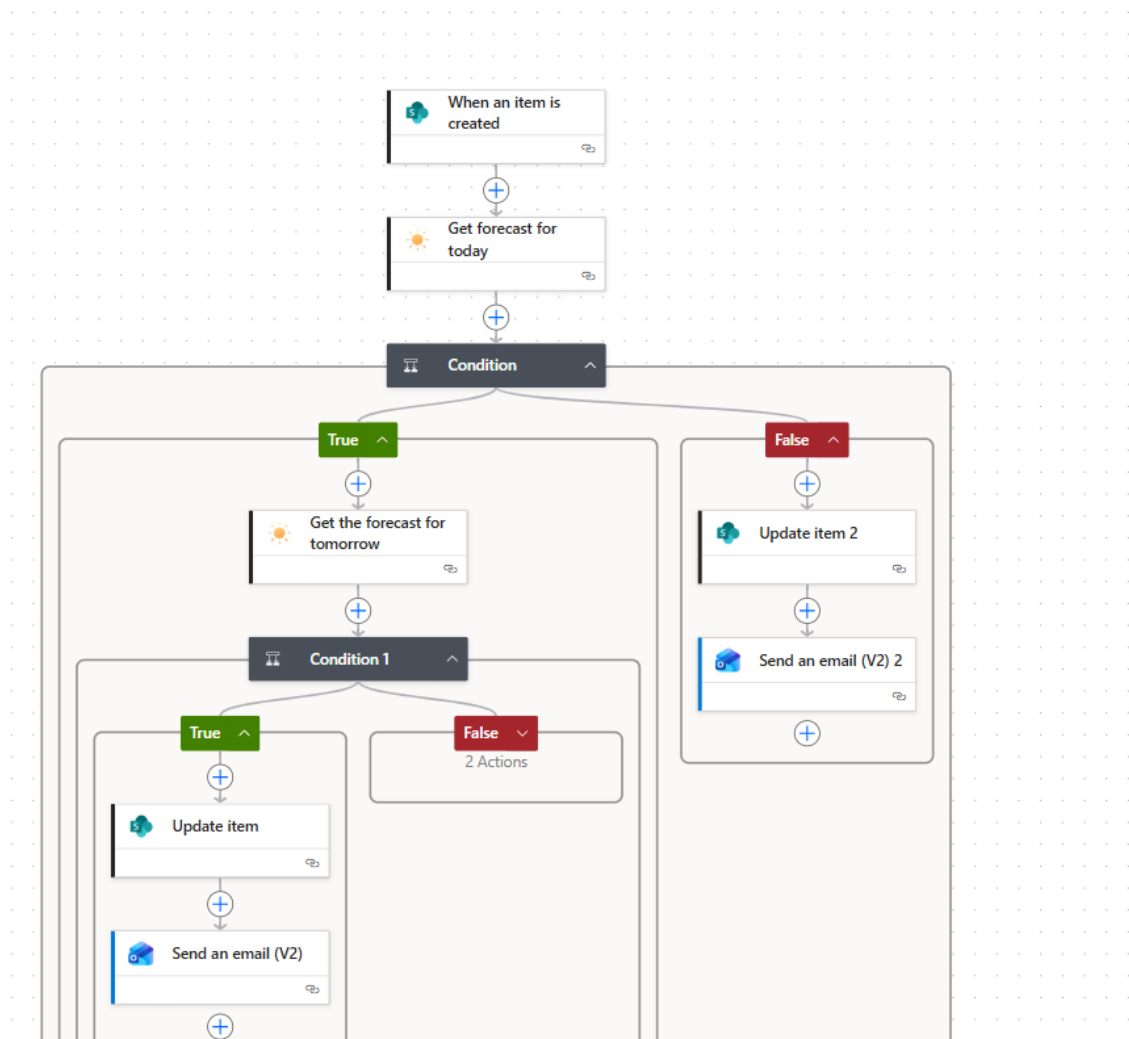


Figure 6 – Power Automate Flow

Along with SharePoint another source of technology used for this prototype was Power Automate, this technology being the most important throughout the whole stage of development. Power Automate allows the data on SharePoint to be processed and added to the list. Components such as get forecast for today and tomorrow are displayed meaning the app will display the forecast for both days and update SharePoint data if the weather for that day is appropriate or needs to be rescheduled. When an item is created through the app it gets the forecast for today and automatically checks tomorrow's forecast before deciding to proceed or not with the activity. This flow has helped build my app and contribute towards user safety by checking multiple forecasts and updating users if they should continue with their activity or not. Power Automate allows automated workflows improving business processes and enhancing user experience (Bhosale, P.G., 2024)

Update item

Parameters Settings Code view Testing About

Advanced parameters

Showing 9 of 11 Show all Clear all

Title

Title X

X

Activity

Activity X

X

Start Time

Start Time X

X

End Time

End Time X

X

Severity Threshold

High

X

Weather Status

Cancelled

X

Weather Reason

Weather conditions are unsuitable today and tomorrow due to rain, snow, wind, storm risk, or freezing temperatures

X

Email

Email X

X

Location

Location X

X

Figure 7 – Update Item

The fields here are directly linked to the column names in the SharePoint list effectively linking the 2 sources together meaning seamless integration. Messages such as high severity and weather reason are manually typed meaning when a user does enter data SharePoint displays the reason for the activity going ahead or not instead of just saying rescheduled or cancelled.

 Update item 1

Parameters

Settings

Code view

Testing

About

Advanced parameters

Showing 9 of 11

Show all

Clear all

Title

 Title X

X

Activity

 Activity X

X

Start Time

 Start Time X

X

End Time

 End Time X

X

Severity Threshold

Medium

X

Weather Status

Rescheduled

X

Weather Reason

Activity rescheduled to tomorrow due to poor weather conditions today

X

Email

 Email X

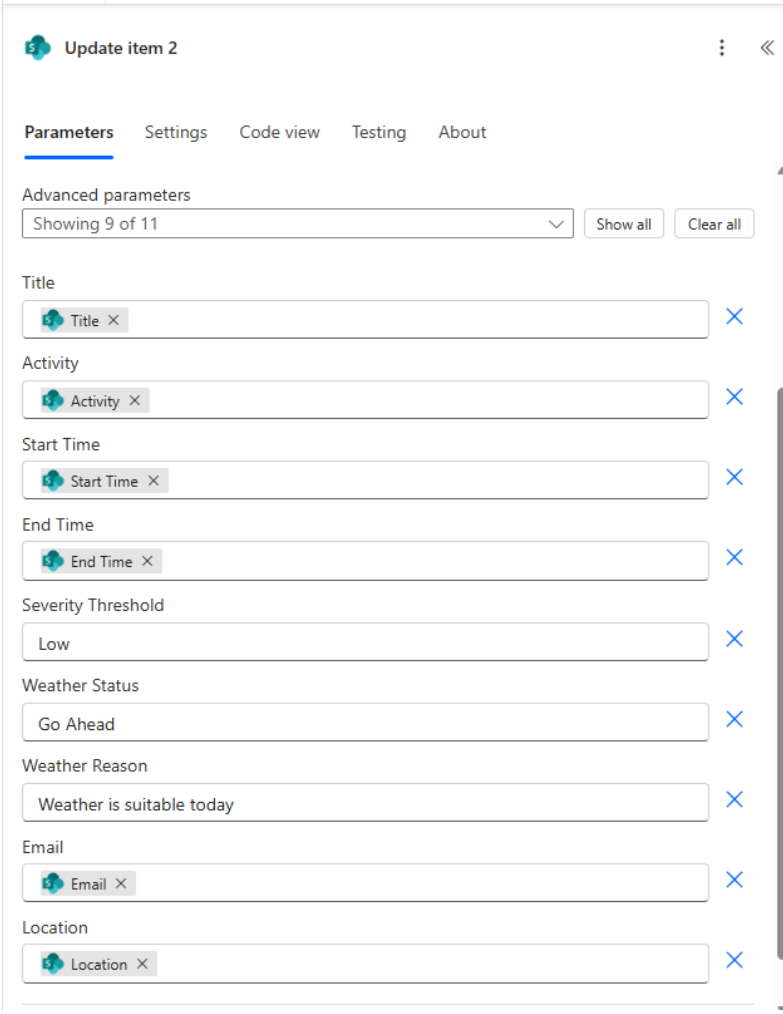
X

Location

 Location X

X

Figure 8 – Update Item False



Update item 2

Parameters Settings Code view Testing About

Advanced parameters

Showing 9 of 11 Show all Clear all

Title

Activity

Start Time

End Time

Severity Threshold

Weather Status

Weather Reason

Email

Location

Figure 9 – Low Severity Item

This screenshot shows the severity based on good weather conditions compared to other 2 items which rescheduled or cancelled based on poor weather displaying reasons why it shouldn't proceed. This one shows the updated item for good weather so when a user enters data through the app if the weather for the city on that day is good the low severity and go-ahead status will be displayed on SharePoint. As a summary based on weather conditions a user can expect 3 types of alerts, rescheduled, cancelled or go ahead.

6. Implementation

6.1 Construction of the Artefact

When a project is being built it needs a clear development process documenting the individual steps and the approach to design. This section will explain the development process in steps

covering multiple details as well as the validation and testing of the design. For my development process I will be breaking down the tools, methods used and other key decisions I have used along the way as well as the challenges encountered during development

Step 1 of the development process included data collection in which responses were gathered clearly stating and identifying problems users had with existing applications and what they wanted to see my app fix over the others. Key decisions were made through this step which led me to identifying tools and software appropriate to solving user needs and enhancing user experience. Initially the challenges I came across during this step involved the number of responses I received, It was a challenge trying to publicly get my form out there for anyone to see which led me to alter my approach and send the form link to professionals including peers which they could access and then answer the questions. In total trying to get a fixed number of responses was a challenge but wasn't necessarily important as I only needed about 6-10 to make key decisions.

Step 2 involved the software's; this is the step where I used languages such as Python for the development within Visual Studio Code. The reason why I chose the hard-coded method first was because of the advantages it had such as full access to customisation, ability to integrate new technologies into it and many more, but the key challenges I had with the coded approach was the reason why I changed and altered my method for the development. I came across many challenges during this step as I found myself stuck and lost in the software many times, making me revert or delete existing files to clean the system and start over. Hard coding does provide many advantages but for my case and the prototype I was doing, I was not confident I could build it efficiently using code especially for the problem I was trying to solve as multiple lines of code would have to be changed, new files would be created and the hardest part integrating external API data into lines of code proved a challenge that was going to take a lot of time to fix and properly navigate which I why I decided to change my approach and keep my coded applications minimal and basic.

Step 3 is the change of approach briefly spoke about towards the end of step 2, Whilst hard coding an app can give you many advantages such as full customization and better access to integration and features it requires a lot of technical expertise and knowledge not to mention how time consuming it is to get every line of code perfect and the go through the debugging process, for me although I have coded versions of apps I decided to switch my approach to something less time consuming and something I enjoyed a bit more than code. This is where Microsoft Power Platform steps in and became the 3rd step into my development as it was a different approach using new tools and technologies. Power Platform tools are less advanced than coded solutions as it provides less customization and data integration than the alternative however, for a prototype this method is perfect especially due to the fact how modern-day businesses are using Power Platform for app development. The lower access to customization and premium features for more in-depth data integration were the main challenges encountered during development but was something I was not fussed about as it still gave me great options to build a technical level app and automate the data in the background which is

something I wouldn't have been able to do through Python or Java programming. The key decision to changing approach came down to time and enjoyment, I felt I was building a basic app through code not advanced enough to present enabling me to change my approach and start from scratch.

Step 4 is where the design using Power Platform approach begins, after identifying the solution in step 3 and changing my approach the development for the prototype begins again. Features such as labels, text inputs and buttons are used along with languages such as Power FX to display location and weather data. This makes it a lot easier than VS Code as the programming language is easier and it still provides expert level technical solutions. Tools such as SharePoint and Lists were used as Microsoft data sources which were directly linked to the canvas app allowing smooth data integration displaying automated data such as pinpoint location coordinates, nearest city in miles and home coordinates through geolocation and GPS integrated code. The challenges in constructing the app included coding the text and buttons allowing for user input and displaying location + weather forecast data, although the coding isn't as technical as a full stack app these were still challenges I came across during development and used research to improve expertise and code the certain text and buttons to automate data based on user text (entering a city) and automatically update their location whenever they move making it not just a technical app, but an app with extensive features relating more to the intended problem.

The artefact meets the problem statement as it provides clear and efficient data to users as well as the geolocation technology which was the big issue relating to safety, as mentioned before no weather apps display a user's pinpoint coordinates making it harder for them to interpret weather data as the data provided would be the weather for the nearest city. That problem there has already been solved and is an innovative solution due it not being done on existing apps before. The scheduling system built into the app relates back to the problem statement of user safety and efficiency in weather apps allowing the system to pull weather data from coordinates and reschedule users' activities through workflows enhancing user safety sending notifications instead of users remaining unsure and risking themselves during weather circumstances. All these techniques so far meet the problem statement as it is providing valuable insight with optional technologies improving user safety through automated schedules for example if raining in Birmingham at 1pm when a user has an activity such as outdoor dog walking at 1pm to 2pm based on the weather data and coordinates the system can automatically cancel and reschedule to another day notifying the user in the process reducing danger so they can plan ahead instead of them manually rescheduling, this alone alerts users as my project said It would and reduces the risk of injury during outdoor activities. .

6.2 Validation and Testing

Throughout the build, the project has been validated and testing meaning that every major or minor tweak made, or new data added is tested straight away by myself to make sure it works or if it gives back accurate data. As I am the developer it is logical for me to test the features and

functionality during the development making sure of accuracy and taking notes of future implementations or strategies to solve testing problems. This approach to testing is a beneficial approach as I can work along the way with features that don't work fixing them throughout the process instead of waiting until the very end of development and having a whole host of problems to solve. The core things that have been validated and tested during this process includes the consistency, functionality and performance. These are the 3 components tested by me and have been documented as test results in tables; this is so it always me to track what works and what doesn't while also knowing what features need to be improved on to achieve high performance, this is an effective approach and strategy to test the core application. Features tested include navigation buttons, switching pages, data entry and updated records. The following tables show my testing documentation of functionality and performance.

Test	Test Result
Switch From page 1 to page 2 through icon and back	Success
Show real-time data for 2 separate forecasts based on user data entry	Success
Update user coordinates automatically, displaying miles from cities	Success
Latitude and Longitude automatic updates	Success
SharePoint data storage through user data live on app	Success
Rescheduled and Cancelled Plans for Users based on weather or conditions	Success
Email Notifications based on rescheduled, appropriate or cancelled activities	Success
Celsius to Fahrenheit	Success
Automatically Empty text input boxes after data entry are saved	Success
Display exact user location	Success
Add activity data with date, start, end and save	Success

The table above shows my self-testing results; results I conducted throughout the process of development. All of the major tests have been a success but all required work and research to get them to work to the way I wanted to, most of these successes were failures and with my approach of fixing them during the development allowed me to leverage the app to a larger scale meaning a functional app and less mistakes towards the end of the app compared to

having multiple functions broken and not working taking the effect of the app down, testing the app along the way makes the process much smoother as less mistakes will be present in the end design contributing to a better application more fitting and user-ready.

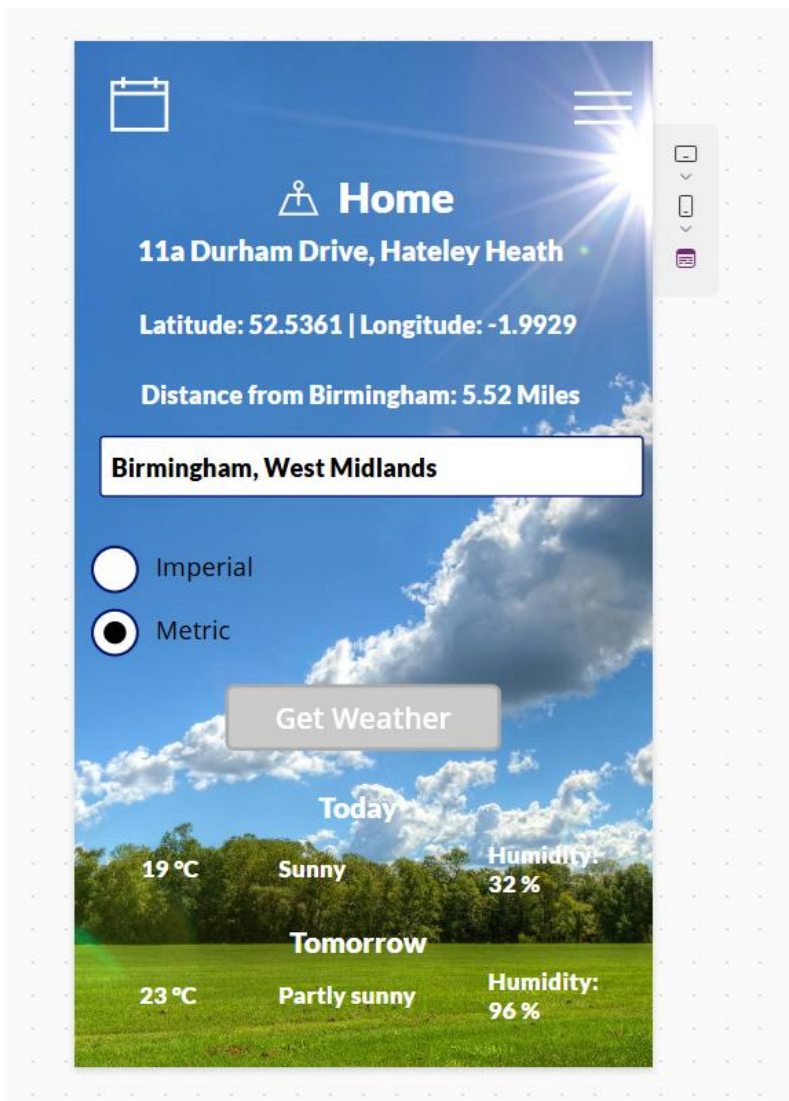
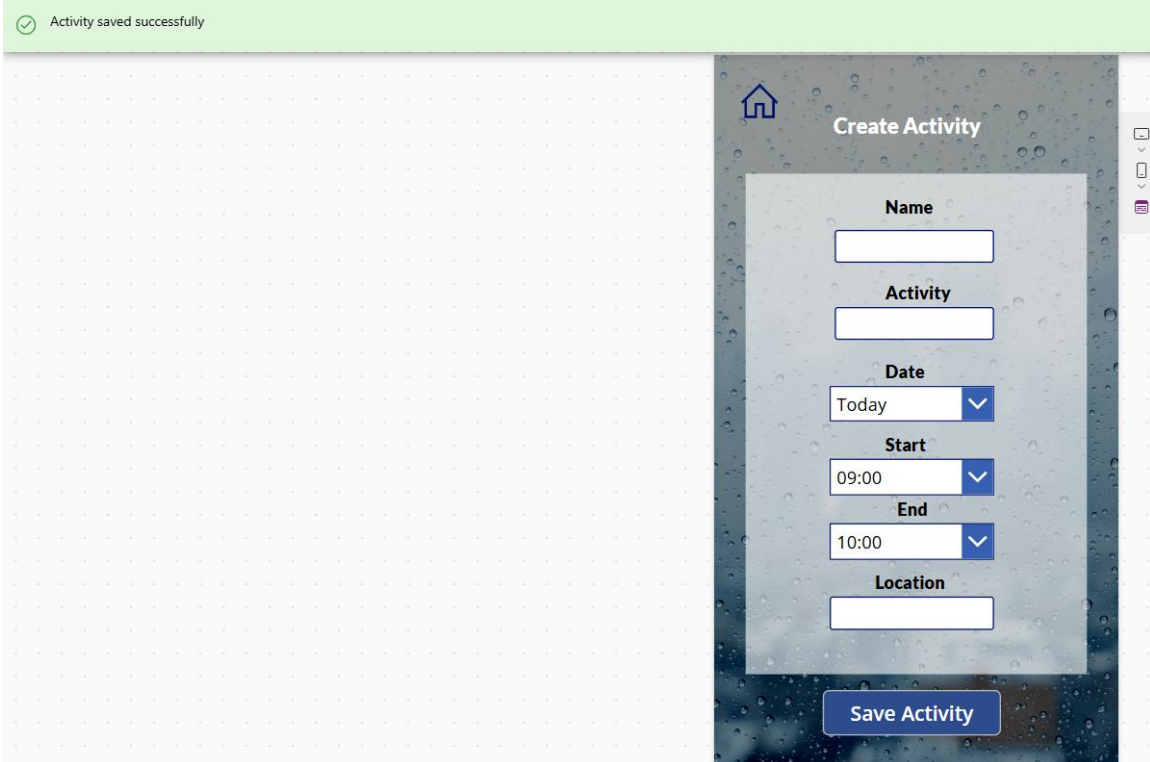


Figure 10 – Home Screen

As mentioned, self-testing was conducted throughout the development of the artefact, and this figure shows exactly what is tested linking back to the previous table. I tested that the automation features work by using the app in multiple locations to check if user coordinates changed or remained the same, if they changed based on movement it was a clear sign of app progression and meant that the automation features worked. User location below Home updated and so did distance from Birmingham. Depending on how far I was from Birmingham the miles would always change here from the screenshot it shows I was 5.52 miles away but when in Birmingham it frequently updated to 0.20 miles showing effective use of automation of

GPS technologies. Get Weather was tested by clicking the button after entering a city with it displaying today's and tomorrow's forecast for that chosen location



Activity saved successfully

Create Activity

Name

Activity

Date

Today

Start

09:00

End

10:00

Location

Save Activity

Figure 11 – Page 2 Saved Activity

Having an activity saved successfully shows testing has been done properly and to a good level as every major component shows evidence of the artefact's logical consistency, functionality and performance just like it does from this screenshot, I type in a user's name and specific activity using the dropdown menu to select a date, start time, end time and location and then saving the activity, this test alone shows the app functions well as the main data is stored in a SharePoint List where all the automation happens.

UserActivities ☆										All Items ▾		+ Add view
Name	Activity	Start Time	End Time	Severity Threshold	Weather Status	Weather Reason	Email	Location		+ Add column		
Randy Turner	Running	09:00	10:00	Medium	Pending	Awaiting forecast	Rohan.Manku@mai...	Yakutsk				
Lisa Turner	Walking	09:00	10:00	Low	Go Ahead	Weather is suitable today	Rohan.Manku@mai...	Yakutsk				
Lisa Turner	Walking	09:00	10:00	Medium	Rescheduled	Activity rescheduled to tomorrow due to poor weather conditions today	Rohan.Manku@mai...	Yakutsk				
Rohan Manku	Football	09:00	12:00	Medium	Pending	Awaiting forecast	Rohan.Manku@mai...	Birmingham				
Rohan Manku	Basketball	09:00	10:00	Low	Go Ahead	Weather is suitable today	Rohan.Manku@mai...	Birmingham				
Rohit Kumar	Running	13:00	14:00	Low	Go Ahead	Weather is suitable today	Rohan.Manku@mai...	Terni				
Viktoria Syzmannski	Football	12:00	14:00	Medium	Rescheduled	Activity rescheduled to tomorrow due to poor weather conditions today	Rohan.Manku@mai...	Yakutsk				
Joao Joao	Futbol	13:00	15:00	Low	Go Ahead	Weather is suitable today	Rohan.Manku@mai...	Brasilia				
Maria Ujkovic	Dog Walking	10:00	11:00	Low	Go Ahead	Weather is suitable today	Rohan.Manku@mai...	Zagreb				
Archie Woodcorner	5K	13:00	16:00	Low	Go Ahead	Weather is suitable today	Rohan.Manku@mai...	London				

Figure 12 – SharePoint Data Testing

Activities added are saved automatically onto a SharePoint list created beforehand, this shows effective testing as it links back to figure 2 and the activity saving successfully. Here you can see the activities with information such as severity, status and reason based on the API data for the weather within the home screen. Depending on the location a user chooses in this case was me for testing it pulls the weather data and can reschedule, cancel or approve an activity to go ahead displaying a valid reason why.

7. Critical Evaluation

The success and significance of my work I would say has been to an industry standard level, while I understand its maybe not the most complex system using a complex coded approach at the end of the day, I still believe it to be a smart innovative solution focusing solely on automation. The reason I would say my artefact is a success because of the many features integrated into the app, such as location automation, GPS tracking and activity scheduling these features alone separate it from other weather apps out there. My artefact essentially blends multiple solutions into 1 focusing on user safety for activities. The app allows users to find weather forecasts for different cities across the world and add activities thy would like to do outdoors with the automation playing a crucial part in identifying the weather for the exact location and notifying the user whether their planned activity should go ahead, be rescheduled or cancelled. The significance of this system is major, because it can contribute towards user safety and reduces the manual approach by users, as this system can automate your activities for you saving less time than if you were to manually reschedule or cancel them. The app has a societal impact as it improves user safety and awareness whilst also being targeted for social people who frequently engage in activities. However, the project also has academic significance

due to its importance and relevance to addressing real-world problems, focusing on research gaps and exploring opportunities not done in the market. Many weather apps are basic and just display relevant information which is all that users needs, but when compared to mine no existing apps integrate my features into one solution, this is because my weather app has locations, activity planning and tracking which is something very uncommon in today's systems meaning my approach is innovative. The advantages my app has compared to competitors is the automation of user location and activity approval, rescheduling and cancelling contributing to user safety and email notifications alerting users if they're activity should go ahead or not. However there does come some disadvantages compared to existing systems, the main one being MSN connector only allows a 2-day forecast being today and tomorrow and most weather apps out there operate with 7–10-day forecasts allowing broader ranges for users to check specific weather data across a week rather than 2 days. This is something that will be worked on in future plans but overall compared to other basic systems my design is quite creative and innovative sparking interest of users due to the unique and unseen features on a weather application.

7.2 Impact

When designing my solution, the problem was clear, to improve user safety and automation, reducing the need for manual checks and I would say my project has had an impact on that topic. The impact it has is a great impact simply because it reduces the need for manual rescheduling with automation and improves user safety and awareness. The project itself addresses a problem for particular communities, people who go outdoors at a lot and partake in outdoor activities, demographics will point out my project to be aimed particularly for elderly or adults in general due to activities such as walking or dog walking being present on the app as they are activities frequently done by them. In general, it is for anyone, athletes, regular people etc, any person who has an outdoor activity planned it is for them not just a specific target audience. With all the functions and technologies implemented into my prototype I believe it to leave a great impact for users being an innovative solution and solving problems not many systems would think to solve but is an ongoing issue. The project brings a new academic discussion because of the implementation and optional features integrated into a weather app, how the multiple features are automated and contribute to user safety and awareness and how the solution differs from other weather apps out there.

8. Conclusion

The journey towards where I am today has been great but rewarding, it started off by collecting user data narrowing down specific problems they have with current applications from their experiences and from there identifying problems in today's world users face with weather apps and how I can make an innovative solution impacting the public. The initial problem was solving the false sense of security problems within weather data through apps but later switched to a better problem focusing on reducing manual input and user safety with forecasting alerts.

Switching problems made it easier for me to develop as the original problem wasn't logical as no weather app would have the most accurate data. The project journey took months and led me to 2 different approaches within the design, approach 1 was full stack coding and approach 2 was automated cloud solutions, changing approach mid design allowed more flexibility and more time to balance out other tasks within development. Many key findings had been discovered along the way including operational efficiency between 2 approaches and customisation freedom between automated cloud solutions and full stack code development. To conclude my reflection of the project's success is innovative, the solution created brings a whole new perception and idea to traditional weather apps due to the use of technologies and features integrated into the system giving it a unique value in today's market.

8.1 Limitations

After review of my chosen methods I can say I believe it to have been the right approach for me during this time even though low code development has its constraints such as less customization and access compared to full stack it makes the development a lot easier and is progressively used more in industries for solutions. Practitioners suggest the LCD platforms lack customization and complained about the loading and publishing of them ([Luo et al., 2021](#)). Other challenges commonly identified with low code included complex issues still needed coding and less powerful than traditional full stack programming ([Luo et al., 2021](#)). These challenges mentioned did make my project a bit difficult and were seen as limitations because of the complex datasets and integrations I used which required critical thinking and programming knowledge to solve therefore making automation processes smoother. Before coding it impacted by project greatly forcing me into research and tools to solve and address the limitation. Another major limitation was the steep learning curve for the platform, as someone who is used to coding platforms such as VS Code and designing basic applications and programs through that, a limitation of low-code was learning the system it had a significant impact at first as was seen as a major limitation but after days of consistency and learning courses I was able to familiarise myself with both Power Apps and Power Automate both of which were significant for my prototype. Even though it wasn't a weakness customization was still seen as a big limitation as full stack coding gives developers the freedom to import and design what they want free of choice with no restrictions, whereas low-code platforms such as Power Apps requires subscriptions to access more features and data models. Overall, most of the limitations were addressed with research and hands-on experience across multiple days and weeks contributing to a ready-made and professional prototype.

8.2 Future Work

The future of the project remains clear while now it's only a prototype in stage 1 there will be plans by me or by other potential investors to continue this project further. Based on the limitations discovered throughout development the plan is clear, change approach to full stack coding with more years of experience, this way it allows full customization and integrations to my app without any charges. Other developers can address the limitations with their experience

and try to alter the app giving it more flexibility and structure for many versions. Limitations could be addressed with low-code solutions and subscription models to enhance data integration and data automation or use full stack coding through programming languages making the application stand out and more technical. If I had more time on the application I would have stuck with my original approach of full stack because of the core advantages it holds over LDC. Front -End frameworks such as React.js provide reusable components, responsive layouts and built-in styling empowering developers to create consistent and visually appealing interfaces. ([Gurusamy and Mohamed., 2020](#)). Similarly, back-end tools such as Node.js provide efficient run time running and storing tasks. Changing the whole approach to full stack unlocks a lot more potential than low-code, while low-code is popular for businesses now due to quick and easy task automation it lacks technical depth compared to full stack which is the plan for future work and models of the existing solution. In terms of future work on Power Apps, alerting systems could be used to give actual warnings like government tests to make sure the users don't miss anything. Below is a table documenting the future of the prototype.

Future Work	Why
Full-Stack Development (React, Node)	Technical depth, better customization and more options available for better interface design and interaction. Better API handling and logic, shows product scalability
Improved weather data handling	Use more reliable API, validate data temperature properly
Multi-Day forecast logic	More efficient planning and schedules, easier decision making for activities based on more weather data
Dashboard & Analytics	Show upcoming activities via graphs or tables on the app
AI Decision Making, Machine Learning algorithms	Predicts cancellations, suggests optimal scheduling times and learns from past outcomes

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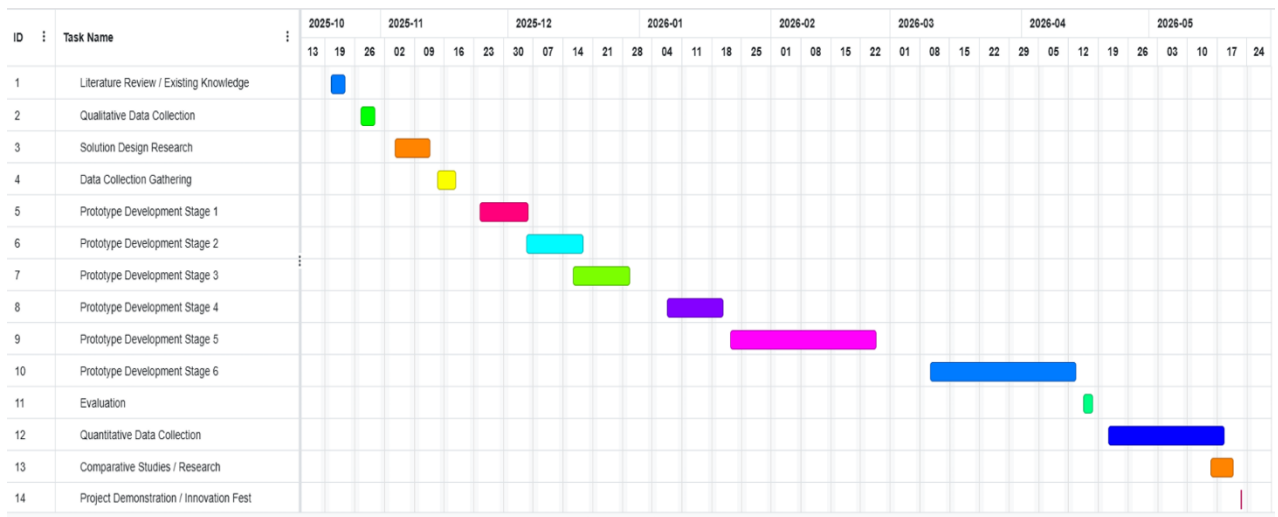
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Appendices

Project Timeline



Feasibility, Risks, and Ethical Aspects

Revised Feasibility

Across 24 weeks my project has been achieved, although there are plans to make it better and more technical it still has been achieved within a time frame. The reason it was achievable was because of the resources I had which allowed me to conduct research on specific technologies and existing software's to develop the prototype, resources such as conference papers, software tools and data collection methods such as forms gave me the platform for this project.

Updated Risk Analysis and Mitigation

Potential Risk	Likelihood (Low/Med/High)	Impact (Low/Med/High)	Mitigation Strategy
<i>Lack of Feedback for Design</i>	<i>Medium</i>	<i>High</i>	<i>Gather enough feedback from users</i>
<i>Project Management</i>	<i>Low</i>	<i>High</i>	<i>Set realistic timelines and objectives</i>
<i>Poor Platform for Designing</i>	<i>Medium</i>	<i>High</i>	<i>Conduct research and test different software's</i>
<i>Scope Creep</i>	<i>Low</i>	<i>High</i>	<i>Create project management framework to manage goals and rigorous changes to development</i>
<i>Data Inaccuracy or Complex Integrations</i>	<i>High</i>	<i>High</i>	<i>Research and navigate alternatives to high end coding</i>

Table 4.1: Risk Assessment and Mitigation

Revised Ethical Considerations

During the course of development, I was able to secure consent from individuals for data collection, having this consent allowed them to give their opinions of struggles they have with application without worrying that their data being exposed as it was all anonymised. The participants each agreed to partake signing a declaration form in the process meaning I have consent to use their data. Having their consent and clarifying their data will be anonymised falls into the government data protection act 2018 which controls how personal information is used by organizations. As the data was used, I followed the guidelines following the information fairly, by not sharing the data with anyone and not keeping their information longer than necessary.